Lab 2 :Prise en main et utilisation de Jenkins

Partie1 : Prérequis : Java 21 et Jenkins

1.Installer Java 21 :

```
arij@master:~/Jenkins-Demo$ sudo apt install -y openjdk-21-jdk
Installation de :
  openjdk-21-jdk

Installation de dépendances :
  openjdk-21-jdk-headless

Paquets suggérés :
  openjdk-21-demo  openjdk-21-source  visualvm

Sommaire :
  Mise à niveau de : 0. Installation de : 2, Supprimé : 0. Non mis à jour : 25
Taille du téléchargement : 84,7 Mo
  Espace nécessaire : 98,2 Mo / 12,3 Go disponible

Réception de : 1 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-21-jdk-headless amd64 21.0
.12.1+1-1~26.04.4 [83,0 MB]
Réception de : 2 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-21-jdk amd64 21.0.12.1+1-1
~26.04.4 [1 656 kB]
84,7 Mo réceptionnés en 27s (3 130 ko/s)
Sélection du paquet openjdk-21-jdk-headless:amd64 précédemment désélectionné.
(Lecture de la base de données... 151026 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../openjdk-21-jdk-headless_21.0.12.1+1-1~26.04.4_amd64.deb ...
Dépaquetage de openjdk-21-jdk-headless:amd64 (21.0.12.1+1-1~26.04.4) ...
Sélection du paquet openjdk-21-jdk:amd64 précédemment désélectionné.
Préparation du dépaquetage de .../openjdk-21-jdk_21.0.12.1+1-1~26.04.4_amd64.deb ...
```

2.Vérification :

```
arij@master:~/Jenkins-Demo$ ls /usr/lib/jvm/
default-java              java-1.21.0-openjdk-amd64  java-17-openjdk-amd64  java-25-openjdk-amd64
java-1.17.0-openjdk-amd64  java-1.25.0-openjdk-amd64  java-21-openjdk-amd64  openjdk-17
arij@master:~/Jenkins-Demo$
```

```
arij@master:~/Jenkins-Demo$ ls /usr/lib/jvm/ | grep -E "17|21|25"
java-1.17.0-openjdk-amd64
java-1.21.0-openjdk-amd64
java-1.25.0-openjdk-amd64
java-17-openjdk-amd64
java-21-openjdk-amd64
java-25-openjdk-amd64
openjdk-17
arij@master:~/Jenkins-Demo$
```

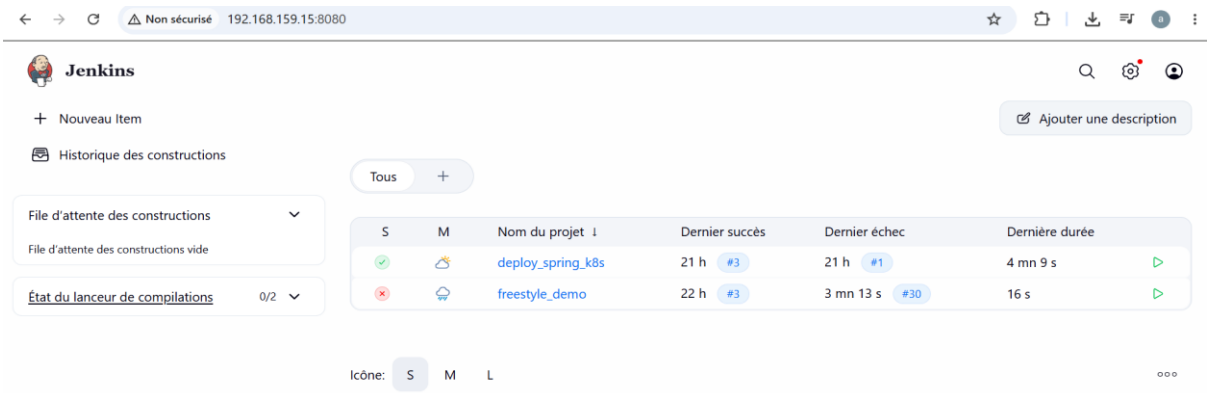
3. Trouver le chemin de Java 21

```
arij@master:~/Jenkins-Demo$ ls /usr/lib/jvm/java-21-openjdk-amd64/bin/javac
/usr/lib/jvm/java-21-openjdk-amd64/bin/javac
arij@master:~/Jenkins-Demo$
```

4. Vérifier Jenkins

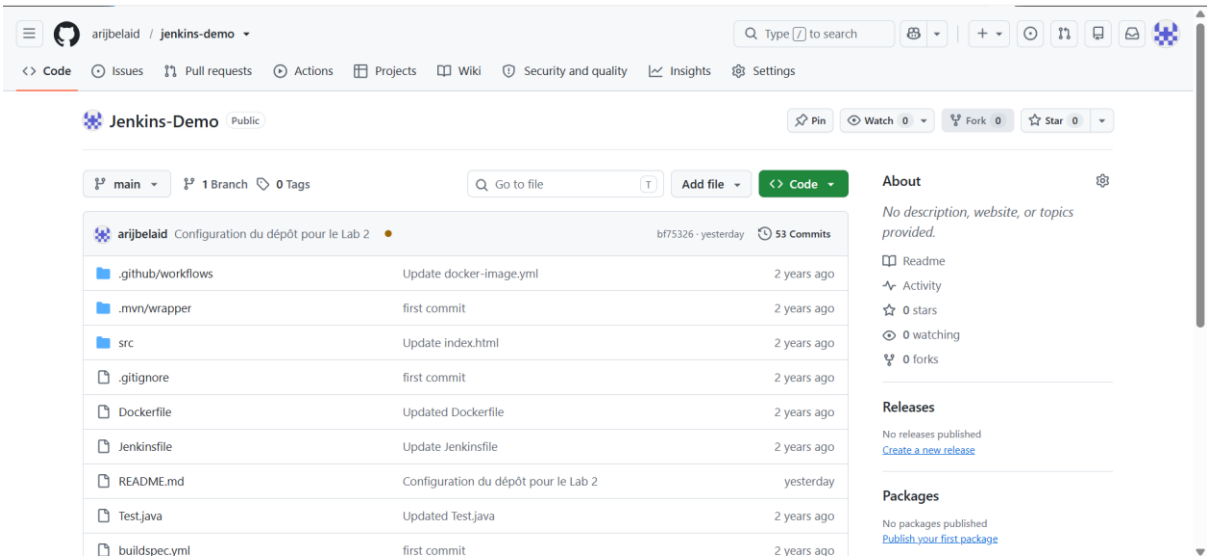
```
arij@master:~/Jenkins-Demo$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-15 19:47:31 UTC; 48min ago
 Invocation: c07dfc7ab0844152a27d90f9cd5ed3e0
   Main PID: 1400 (java)
      Tasks: 53 (limit: 3937)
     Memory: 811.9M (peak: 835.3M)
        CPU: 1min 8.660s
     CGroup: /system.slice/jenkins.service
             └─1400 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenki>
```

5. Vérifier l'accès de Jenkins depuis Windows



Partie 2 : Préparation du projet (Fork + Clone) :

1.vérifier la création du dépôt GitHub :



2.clone du projet sur vm master :

```
arij@master:~/Jenkins-Demo$ ls -la
total 88
drwxrwxr-x  7 arij arij 4096 Oct 14 08:40 .
drwxr-x--- 10 arij arij 4096 Oct 15 20:29 ..
drwxrwxr-x  7 arij arij 4096 Oct 15 20:29 .git
drwxrwxr-x  3 arij arij 4096 Oct 13 22:03 .github
-rw-rw-r--  1 arij arij  303 Oct 13 22:03 .gitignore
drwxrwxr-x  3 arij arij 4096 Oct 13 22:03 .mvn
-rw-rw-r--  1 arij arij  111 Oct 13 22:04 Dockerfile
-rw-rw-r--  1 arij arij 3016 Oct 13 22:03 Jenkinsfile
-rw-rw-r--  1 arij arij 3072 Oct 13 22:03 README.md
-rw-rw-r--  1 arij arij  129 Oct 13 22:03 Test.java
-rw-rw-r--  1 arij arij  737 Oct 13 22:03 buildspec.yml
-rw-rw-r--  1 arij arij  365 Oct 14 08:39 deployment.yaml
-rw-rw-r--  1 arij arij  628 Oct 13 22:03 k8s.yml
-rw-rw-r--  1 arij arij 9114 Oct 13 22:03 mvnw
-rw-rw-r--  1 arij arij 5811 Oct 13 22:03 mvnw.cmd
-rw-rw-r--  1 arij arij 2533 Oct 13 22:03 pom.xml
-rw-rw-r--  1 arij arij  184 Oct 14 08:40 service.yaml
drwxrwxr-x  4 arij arij 4096 Oct 13 22:03 src
drwxrwxr-x  9 arij arij 4096 Oct 13 22:09 target
arij@master:~/Jenkins-Demo$
```

3.Vérifier remote :

```
arij@master:~/Jenkins-Demo$ git remote -v
origin  https://github.com/arijbelaid/Jenkins-Demo.git (fetch)
origin  https://github.com/arijbelaid/Jenkins-Demo.git (push)
arij@master:~/Jenkins-Demo$
```

4.modifier le contenu de index.html

```
 GNU nano 8.7.1                          src/main/resources/static/index.html *
<!DOCTYPE html>
<html>
  <head>
    <title>Arij Belaid - Jenkins CI/CD Pipeline</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        margin: 0;
        padding: 0;
        min-height: 100vh;
      }

      .container {
        text-align: center;
        padding: 60px 20px;
        color: #fff;
      }

      h1 {
        text-align: center;
        color: #fff;
        margin-top: 50px;

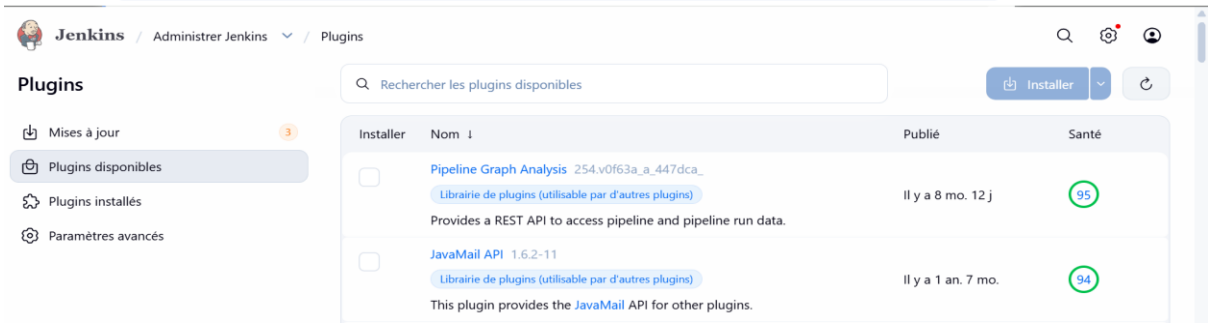
^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo
```

5. Pousser la modification sur GitHub :

```
arij@master:~/Jenkins-Demo$ git push origin main
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 2 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 2.35 KiB | 802.00 KiB/s, done.
Total 12 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:arijbelaid/Jenkins-Demo.git
To github.com:arijbelaid/jenkins-demo.git
   bf75326..a9f4875  main -> main
arij@master:~/Jenkins-Demo$
```

PARTIE 3 — Installation des plugins Jenkins

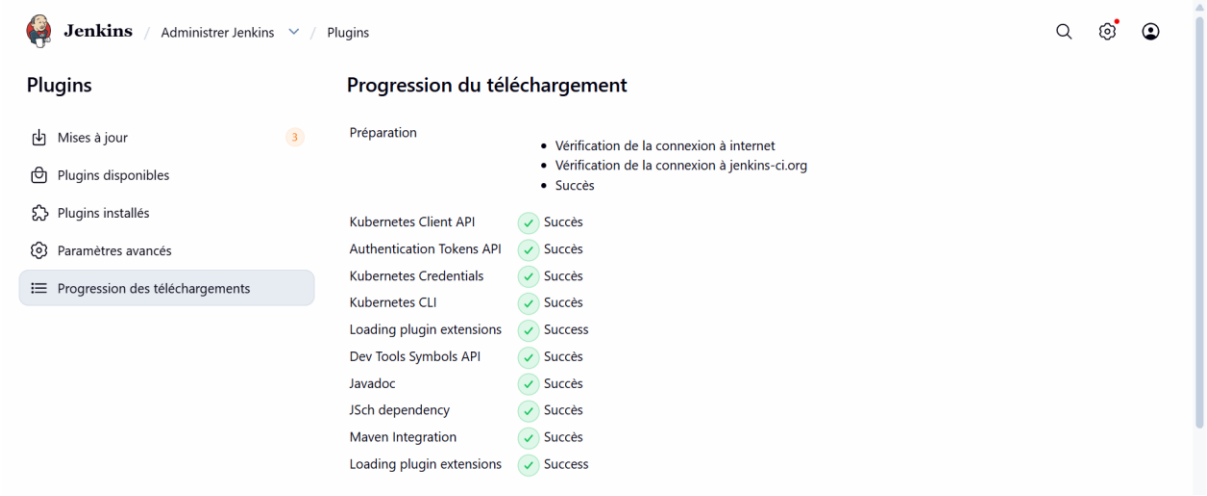
1. Accéder au gestionnaire de plugins



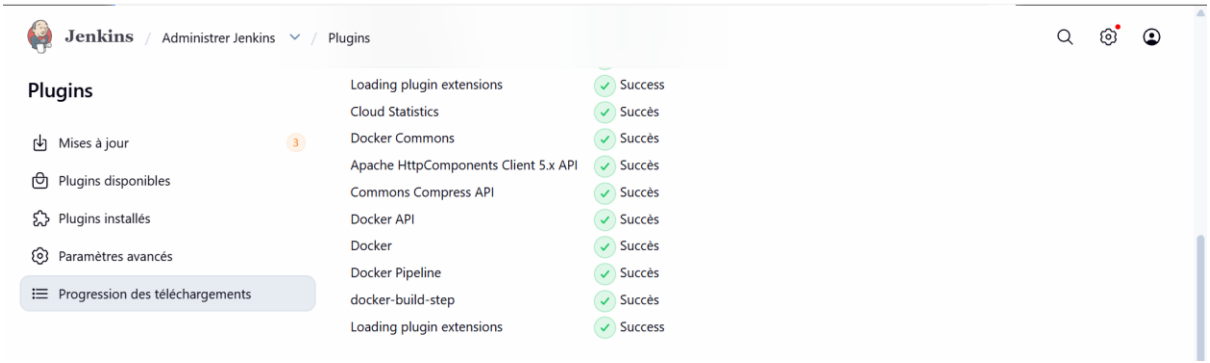
2. Installer le plugin "Kubernetes CLI"



3. Installer Maven Integration

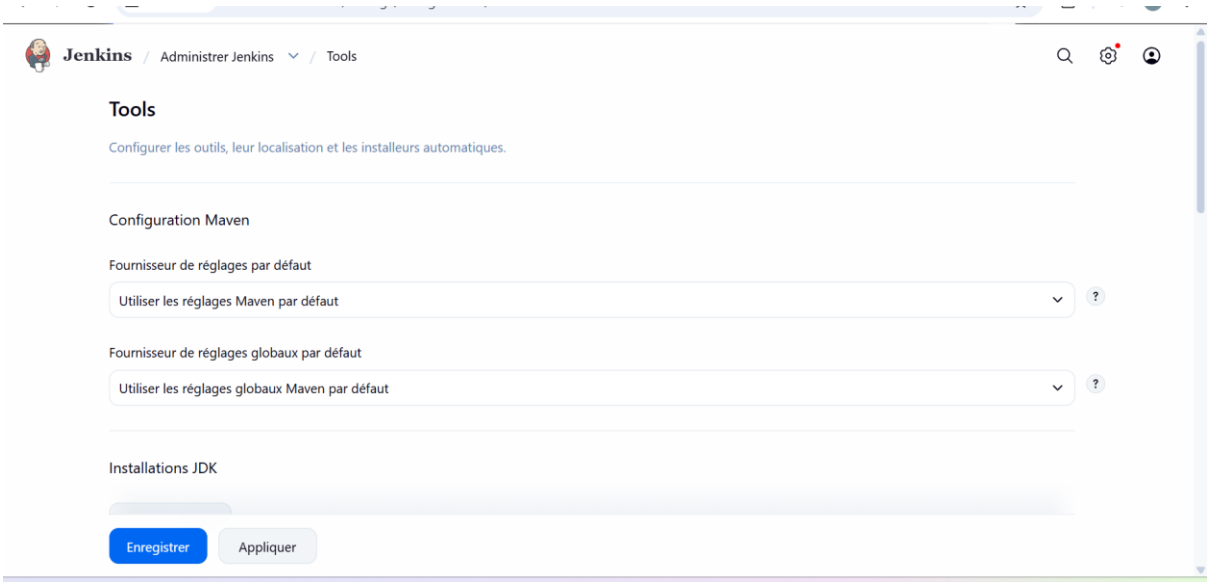


4. installer les plugins Docker :

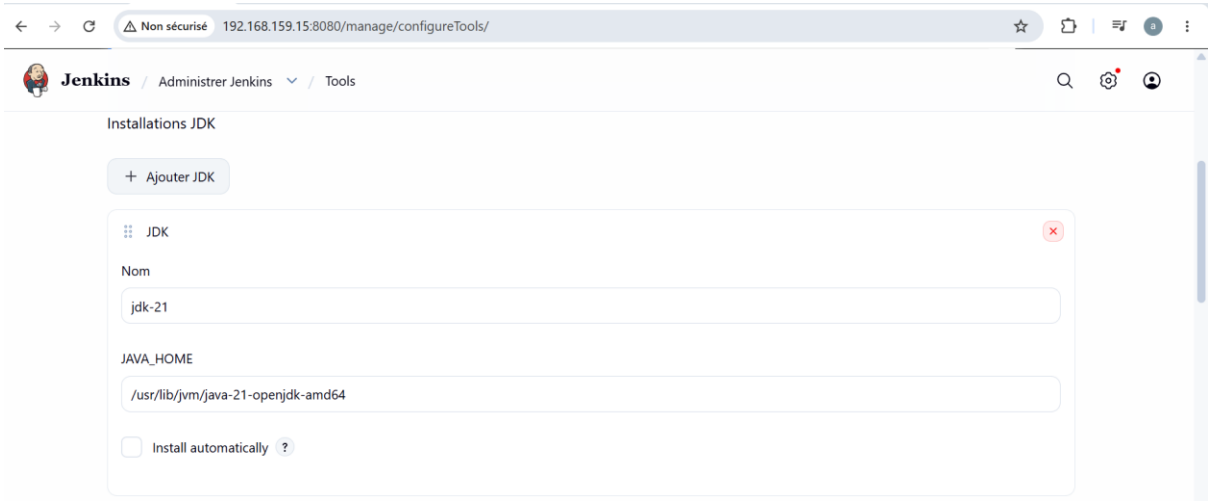


PARTIE 4 — Configuration globale des outils (JDK, Maven, Docker) :

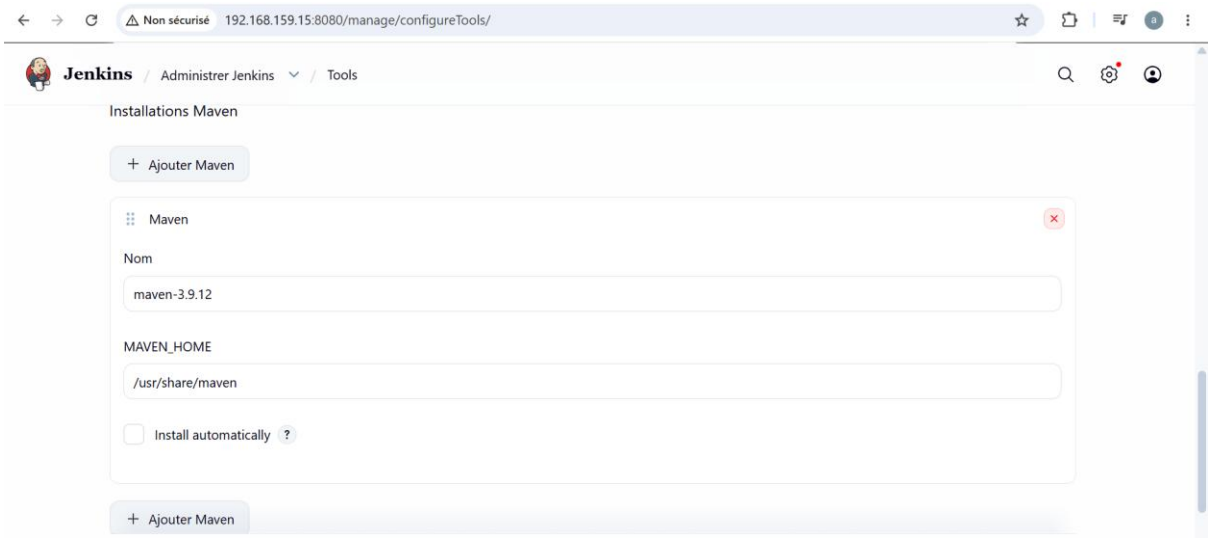
1. Accéder à la configuration :



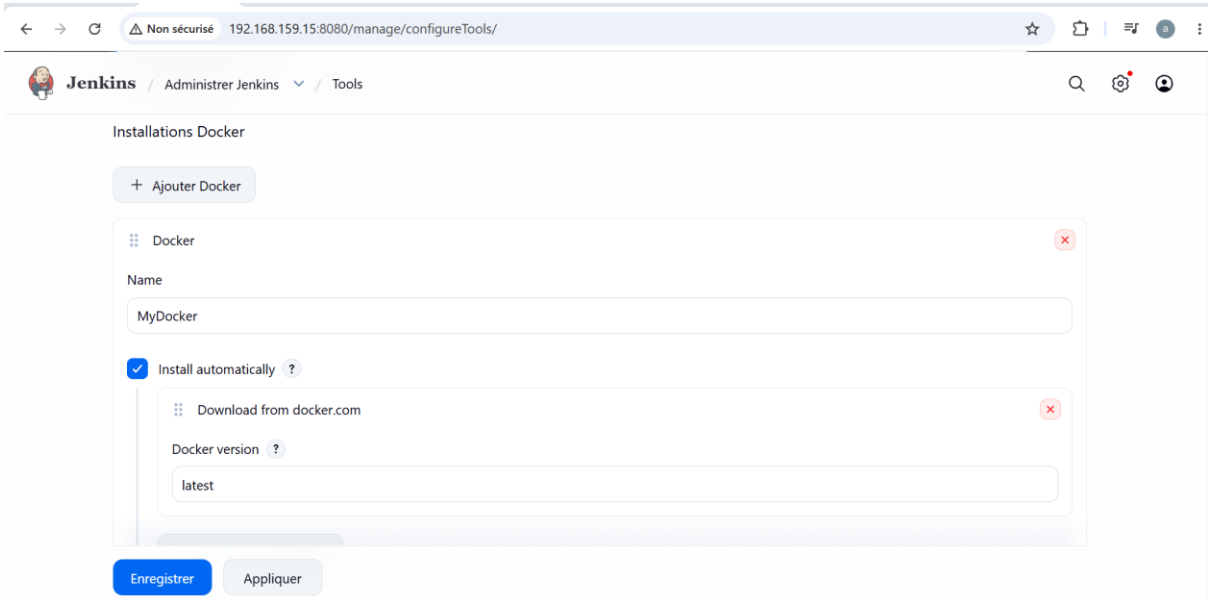
2. Configurer le JDK :



3. Configurer Maven :

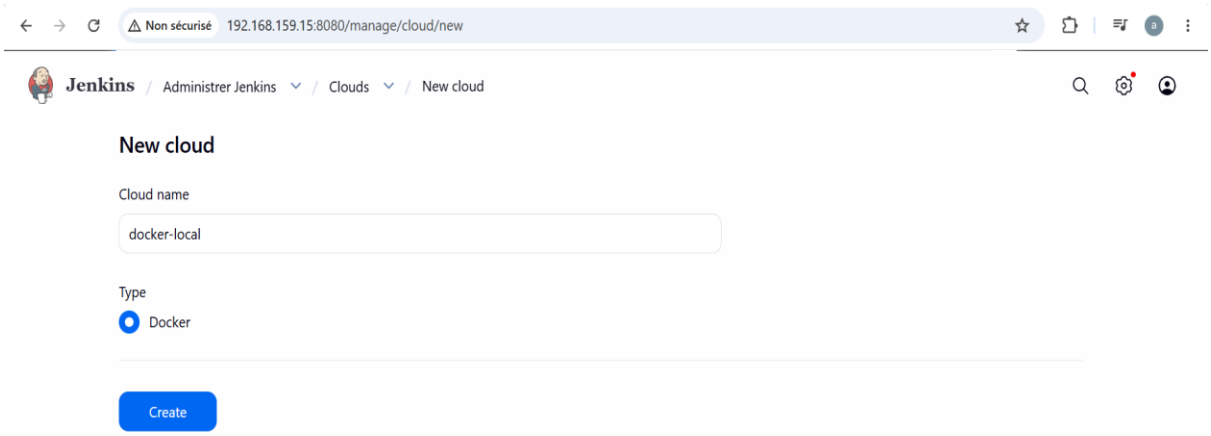


4. Configurer Docker :

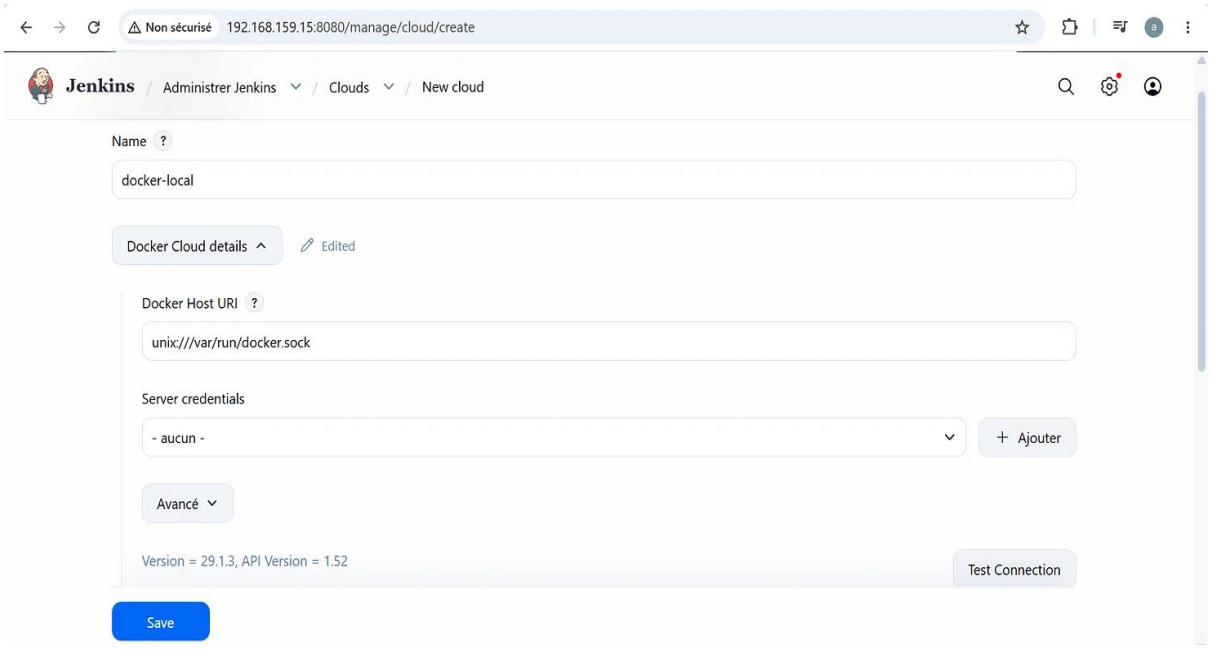


PARTIE 5 — Configuration du Cloud Docker :

1. créer Cloud Docker :



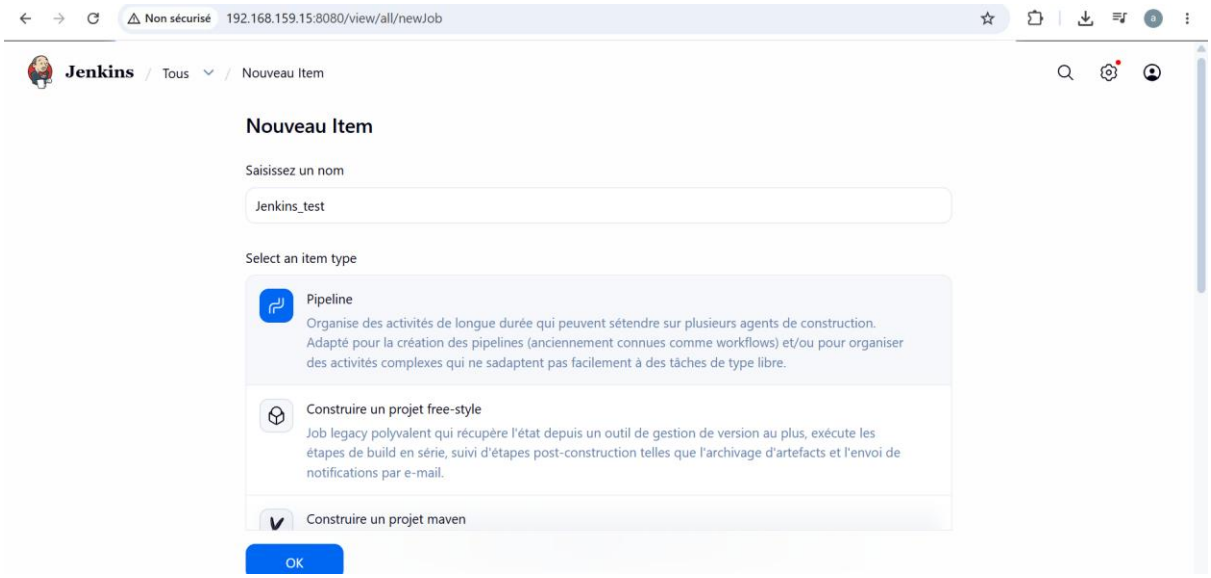
2. Configurer le Cloud Docker



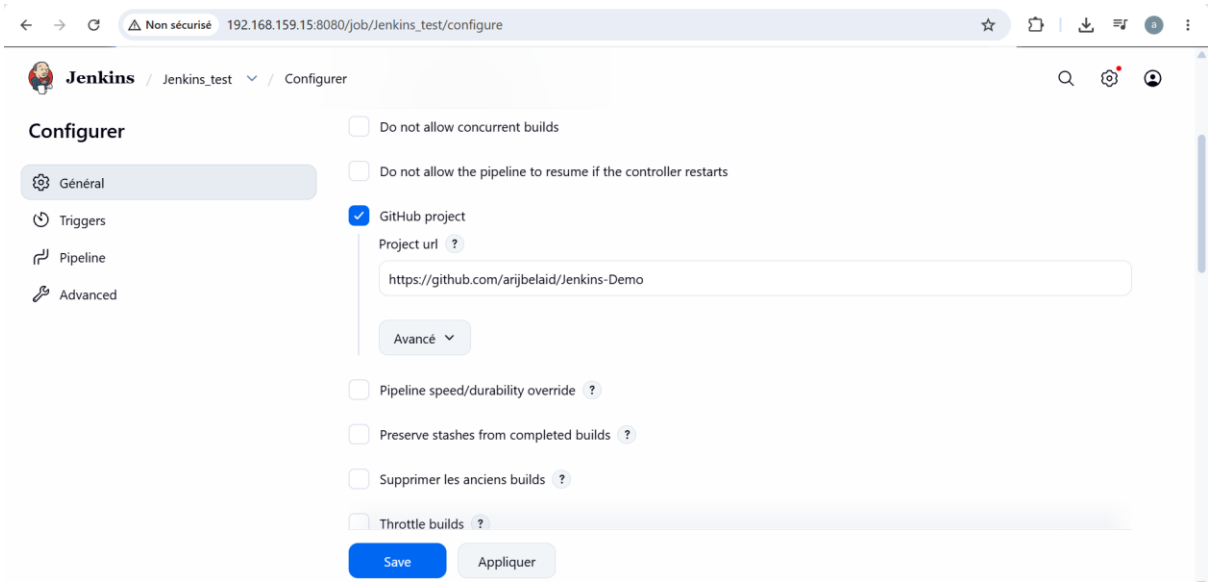


*PARTIE 6 — Création du Pipeline Jenkins**

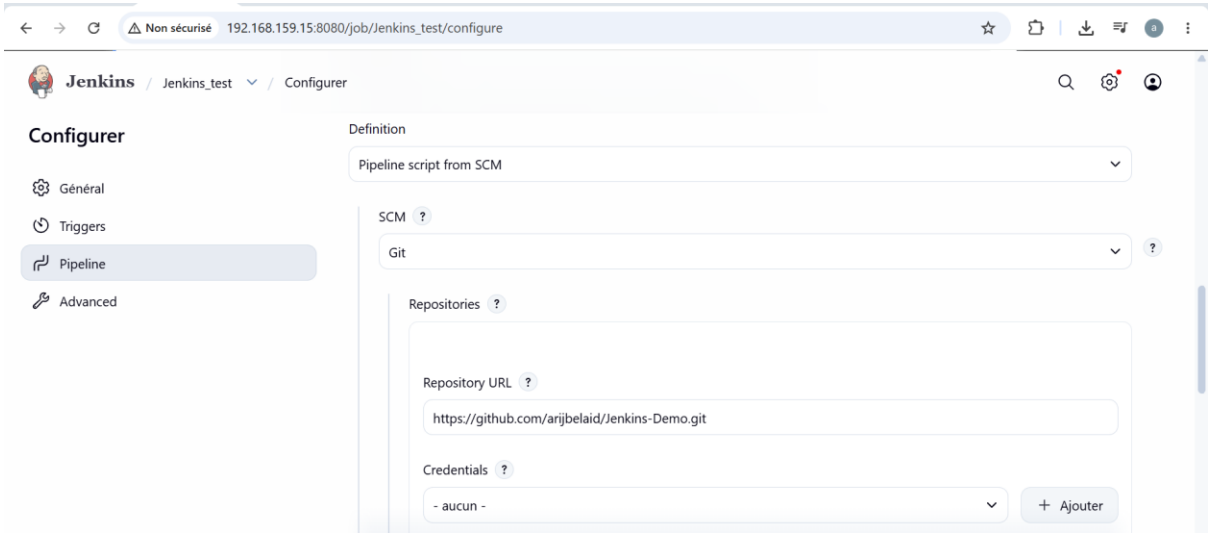
1. Créer le nouveau job



2. Configuration :

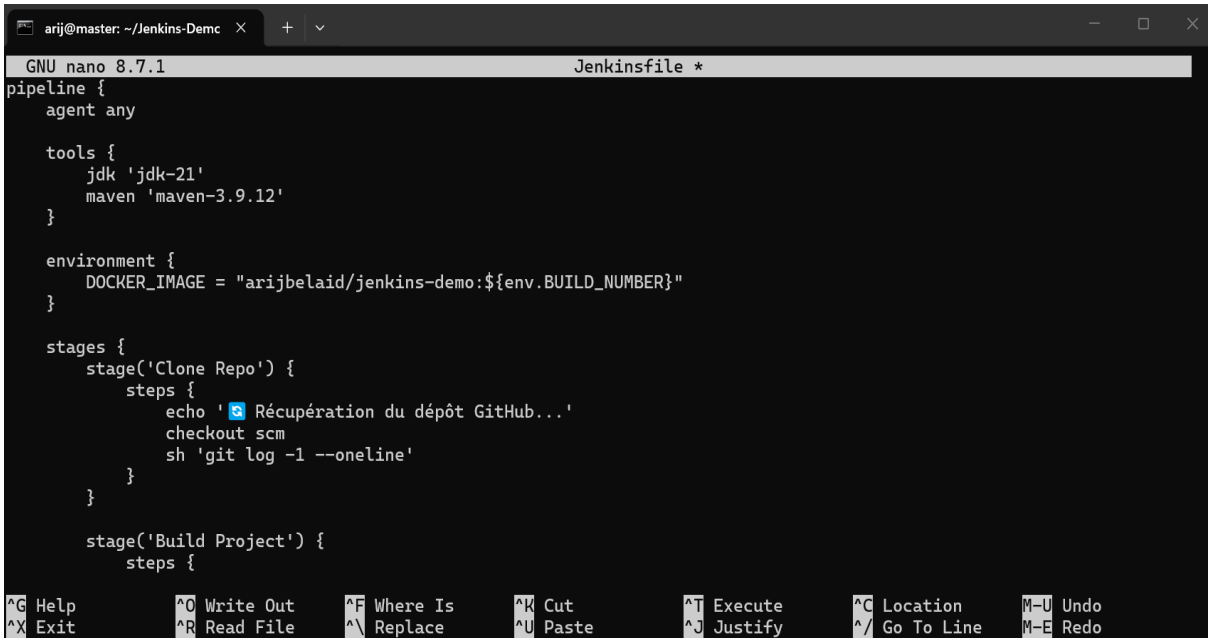


3. Définir le Pipeline depuis SCM



PARTIE 7 — Modification du Jenkinsfile

1. Créer le nouveau Jenkinsfile



2.pousser sur github :

```
arij@master:~/Jenkins-Demo$ git add Jenkinsfile
arij@master:~/Jenkins-Demo$ git commit -m "feat: Jenkinsfile personnalisé pour le Lab 2"
[main 01ddc80] feat: Jenkinsfile personnalisé pour le Lab 2
 1 file changed, 58 insertions(+), 72 deletions(-)
arij@master:~/Jenkins-Demo$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.07 KiB | 274.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaid/Jenkins-Demo.git
   a9f4875..01ddc80  main -> main
arij@master:~/Jenkins-Demo$
```

PARTIE 8 — Lancement du Pipeline Jenkins :

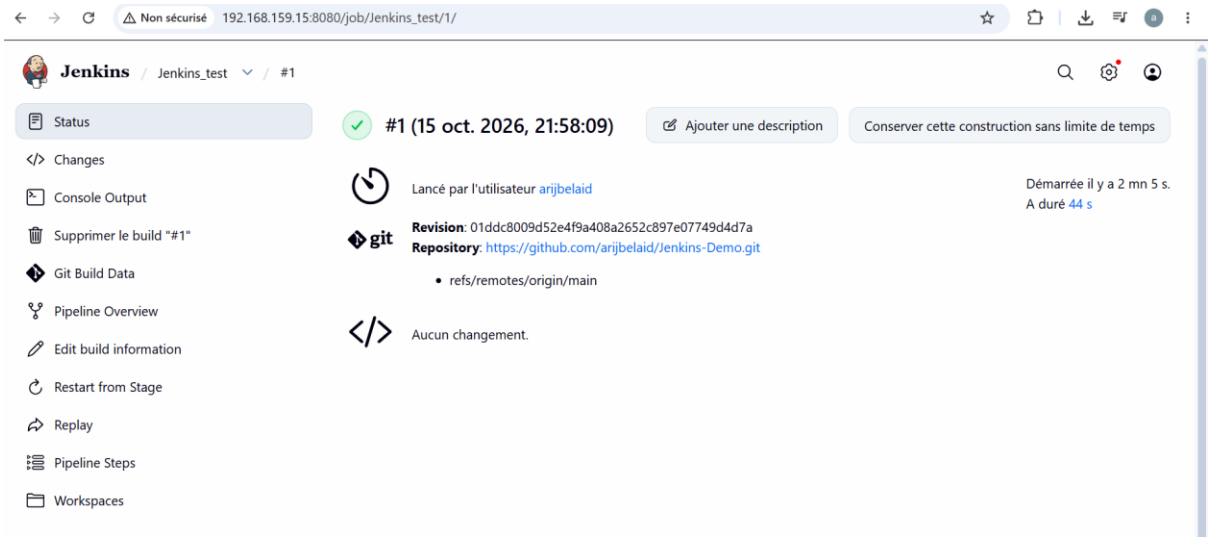
1.vérification sur master :

```
arij@master:~/Jenkins-Demo$ # Vérifier que le port 2222 est libre
sudo lsof -i :2222

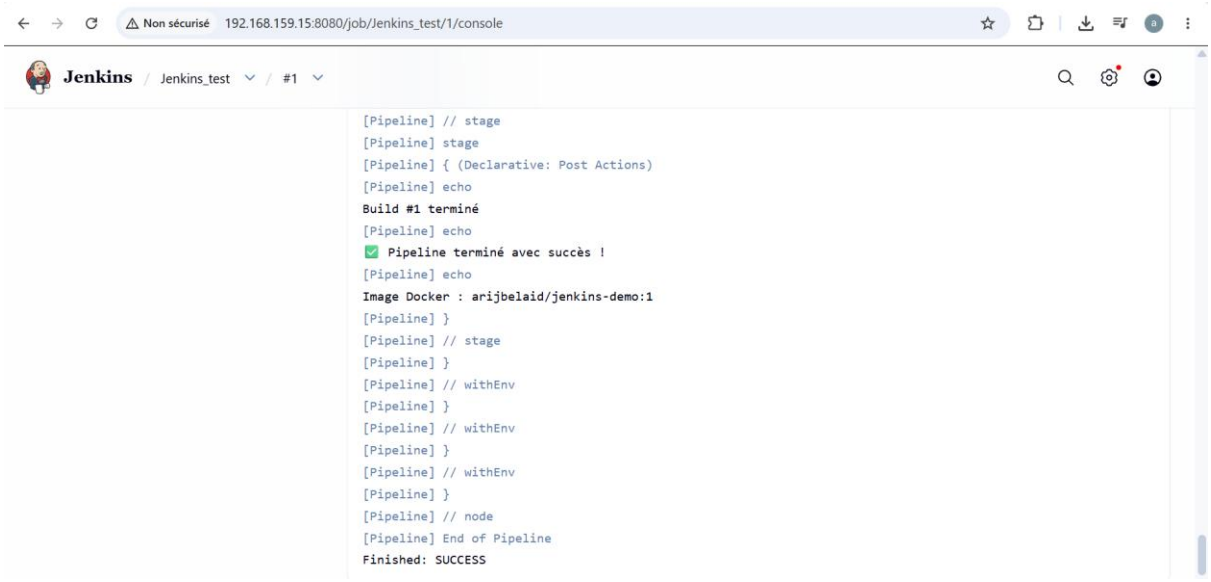
# Vérifier que jenkins peut utiliser docker
sudo -u jenkins docker ps

# Vérifier le Jenkinsfile sur GitHub
cd ~/Jenkins-Demo
git log -1 --oneline -- Jenkinsfile
[sudo: authenticate] Password:
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
01ddc80 (HEAD -> main, origin/main, origin/HEAD) feat: Jenkinsfile personnalisé pour le Lab 2
arij@master:~/Jenkins-Demo$
```

2. Lancer le build :



3. Suivre l'exécution



Vérifier les stages exécutés :

```
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Docker Container)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] echo
```

```
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Fix Dockerfile)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] echo
```

4.Vérification sur master :

```
arij@master:~/Jenkins-Demo$ docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS         PORTS
                  NAMES
bb13566f12a0   arijbelaid/jenkins-demo:1   "/__cacert_entrypoin…"   8 minutes ago   Up 8 minutes   0.0.0.0:2222->2222/tcp, [::]:2222-
>2222/tcp, 8080/tcp   devopsexample
arij@master:~/Jenkins-Demo$ docker images | grep jenkins-demo
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
arijbelaid/jenkins-demo:1        a07c24a73a02        483MB          134MB   U
arijbelaid/jenkins-demo:1.0.0    5fec75671049        483MB          134MB   U
arijbelaid/jenkins-demo:latest   752c0ad2baa2        483MB          134MB
jenkins-demo:1.0.0               5fec75671049        483MB          134MB   U
arij@master:~/Jenkins-Demo$ curl http://localhost:2222 | head -20
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
100   3152 100   3152   0      0 144.1k      0                              0
<!DOCTYPE html>
<html>
  <head>
    <title>Arij Belaid - Jenkins CI/CD Pipeline</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        margin: 0;
        padding: 0;
        min-height: 100vh;
      }

      .container {
        text-align: center;
        padding: 60px 20px;
        color: #fff;
      }
```

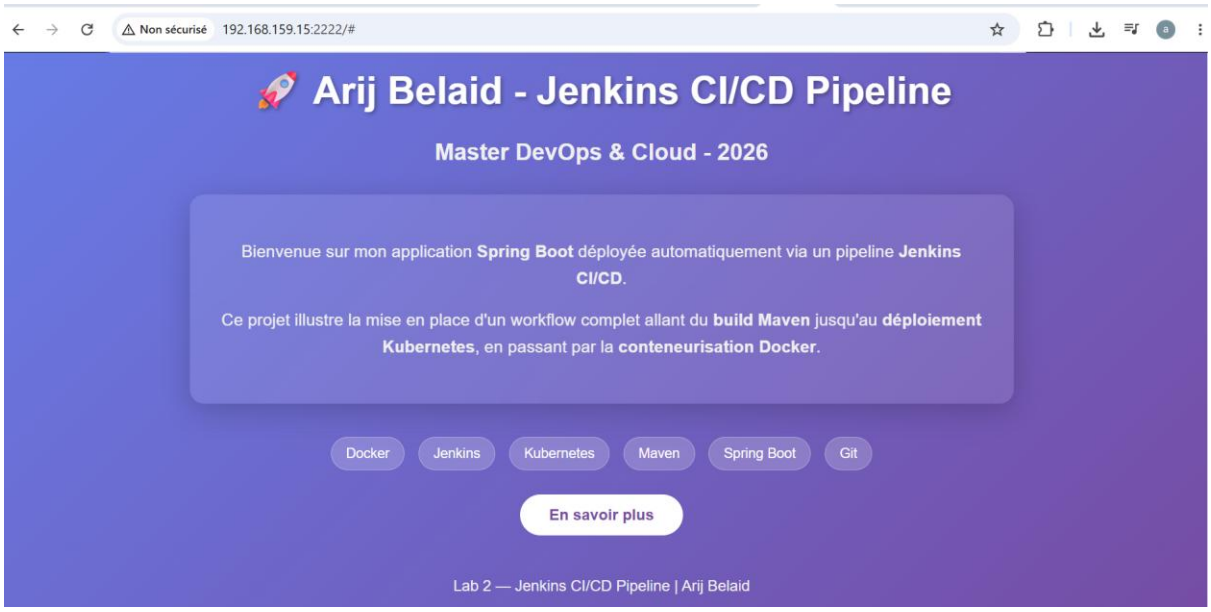
5. Sortie de curl :

```
arij@master:~/Jenkins-Demo$ curl http://localhost:2222 | head -20
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
100   3152 100   3152   0      0 117.6k      0                              0
<!DOCTYPE html>
<html>
  <head>
    <title>Arij Belaid - Jenkins CI/CD Pipeline</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        margin: 0;
        padding: 0;
        min-height: 100vh;
      }

      .container {
        text-align: center;
        padding: 60px 20px;
        color: #fff;
      }

      h1 {
```

6.Tester l'application dans le navigateur Windows


Non sécurisé
192.168.159.15:2222/#
🚀 Arij Belaid - Jenkins CI/CD Pipeline
Master DevOps & Cloud - 2026
Bienvenue sur mon application Spring Boot déployée automatiquement via un pipeline Jenkins CI/CD.
Ce projet illustre la mise en place d'un workflow complet allant du build Maven jusqu'au déploiement Kubernetes, en passant par la conteneurisation Docker.
Docker
Jenkins
Kubernetes
Maven
Spring Boot
Git
En savoir plus
Lab 2 — Jenkins CI/CD Pipeline | Arij Belaid